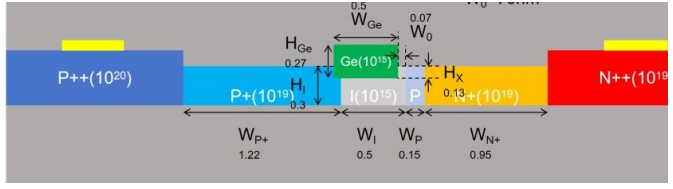
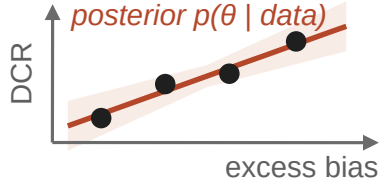
Robust design game: $\max_{x} \min_{\delta \in \Delta} \rho(x, \delta)$ subject to a TCAD budget $\Sigma n_\ell c_\ell \le B$ *designer maximises over designs x; challenger minimises over uncertainties δ*

1 Physics and data engine

WP1 · M1–4

Automated Silvaco ATLAS (team TCAD)





Calibration data

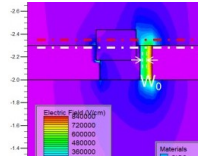
- HWU SPAD data
- published APDs

Uncertainty set Δ: physics θ · process p (e.g. ±20 nm W_P) · bias drift ΔV

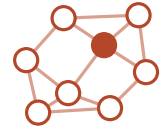
Automate 2D/3D TCAD and calibrate it to measured data → uncertainty set Δ

2 Graph surrogate, robust designer and challenger

WP2 · M3–10



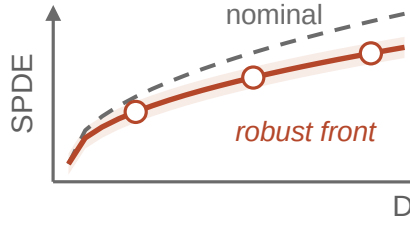
|E| map (team TCAD)



GNN ensemble

SPDE, DCR, g
μ ± σ

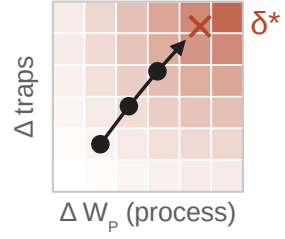
Designer (max over x)



x*

δ*, ρ

Challenger (min over δ)



P_{fail} = Pr[ρ < 0] (share of Δ where the design fails), by Monte Carlo on the surrogate; top δ re-run in TCAD; final designs checked at 250–295 K*

A GNN predicts fields and SPDE/DCR with uncertainty; the designer and challenger play Eq. (2), and top counterexamples are re-simulated

3 Explain, validate and deliver

WP3 · M8–12

- SHAP + symbolic regression → rules
- Blind 3D TCAD validation
- Si test diodes (HWU cleanroom)
- Open toolkit + GeSPAD-Bench data
- Impact: fab roadmap, quantum networks

Turn the surrogate into design rules, validate blind and on test diodes, release the toolkit